

40.520: Stochastic Models**Assignment 3***Due date: Mar 25th, 2026 before midnight*

Problem 1. If X is exponential with rate λ , show that $Y = \lfloor X \rfloor + 1$ is geometric with parameter $p = 1 - e^{-\lambda}$, where $\lfloor X \rfloor$ is the largest integer less than or equal to X .

Problem 2. Events occur according to a Poisson process with rate λ . Each time an event occurs, we must decide whether or not to stop, with our objective being to stop at the last event to occur prior to some specified time T , where $T > 1/\lambda$. That is, if an event occurs at time t , $0 \leq t \leq T$, and we decide to stop, then we win if there are no additional events by time T , and we lose otherwise. If we do not stop when an event occurs and no additional events occur by time T , then we lose. Also, if no events occur by time T , then we lose. Consider the strategy that stops at the first event to occur after some fixed time s , $0 \leq s \leq T$.

- (a) Using this strategy, what is the probability of winning?
- (b) What value of s maximizes the probability of winning?
- (c) Show that one's probability of winning when using the preceding strategy with the value of s specified in part (b) is $1/e$.

Problem 3. A new malware is spreading on the Internet! Our goal is to estimate its spread/damage by time t , assuming it starts at time 0. Assume that internet hosts get infected by this malware according to a Poisson process with rate λ , where λ is **unknown**. Thrasyvoulos installs a Honeypot security system to detect whether hosts are infected. Unfortunately, there is a **lag time** between when a computer is infected by the malware and the Honeypot detects the damage. Assume that the lag time is distributed $\text{Exp}(\mu)$. Suppose that Thrasyvoulos's Honeypot system has detected $N_1(t)$ infected hosts by time t . Thrasyvoulos worries that, because of the lag, the number of infected hosts is actually much higher than $N_1(t)$. The goal of the problem is to understand how many additional hosts, $N_2(t)$, are expected to also be infected at time t .

- (a) Suppose that an infection happens at time s , where $0 < s < t$. What is the probability that the infection is detected at time t ?
- (b) Consider an arbitrary infection that happens before time t . What is the (unconditional) probability p that the infection is detected by the Honeypot by time t ?
- (c) How can we use this knowledge to estimate λ ? (hint: consider using Poisson splitting)
- (d) Use your estimate of λ to determine the expected value of $N_2(t)$.

Problem 4. Let $\{N(t), t \geq 0\}$ be a Poisson process with rate λ that is independent of the sequence $X_1, X_2, \dots$ of independent and identically distributed random variables with mean μ and variance σ^2 . Find

$$\text{Cov} \left(N(t), \sum_{i=1}^{N(t)} X_i \right)$$

Problem 5. Emails arrive at a server according to a non-homogeneous Poisson process with intensity function

$$\lambda(t) = 5 + 2t, \quad t \geq 0$$

where t is measured in hours.

- (a) What is the expected number of emails that arrive between $t = 1$ and $t = 4$?
- (b) What is the probability that exactly 10 emails arrive between $t = 2$ and $t = 3$?
- (c) Given that 50 emails arrived between $t = 0$ and $t = 5$, what is the probability that at least 20 of them arrived after $t = 3$?

Problem 6. An insurance company receives claims according to a Poisson process with rate $\lambda = 5$ claims per day. The claim amounts $X_1, X_2, \dots$ are independent and identically distributed random variables, independent of the claim arrival process, with mean $\mu = \$100$ and variance $\sigma^2 = \$50^2$. Let $S(t) = \sum_{i=1}^{N(t)} X_i$ be the total claim amount up to time t (in days).

- (a) Find the expected total claim amount $E[S(t)]$ and variance $\text{Var}(S(t))$.
- (b) What is the probability that no claims occur in a given day?
- (c) Given that 10 claims occurred in a day, what is the expected total claim amount for that day?
- (d) Suppose the company starts with \$10,000 at time $t = 0$ and earns interest at a continuous rate such that its capital at time t is $C(t) = 10000e^{0.05t} - S(t)$. Find the expected capital at time $t = 30$ days.

Problem 7. An insurance company models the number of claims $N(t)$ arriving by time t (in days) as a **mixed Poisson process** where the claim rate Λ is random. Suppose:

- Λ takes only two possible values:

$$\Lambda = \begin{cases} 2 \text{ claims/day} & \text{with probability } \frac{1}{2} \\ 6 \text{ claims/day} & \text{with probability } \frac{1}{2} \end{cases}$$

- Given $\Lambda = \lambda$, claims arrive as a Poisson process with rate λ .

- (a) Find $E[N(t)]$ and $\text{Var}(N(t))$.
- (b) What is $P(N(1) = 0)$, the probability of no claims in the first day?
- (c) Suppose we observe that $N(1) = 4$. What is the conditional probability that $\Lambda = 6$?
- (d) Using part (c), find $E[\Lambda \mid N(1) = 4]$.